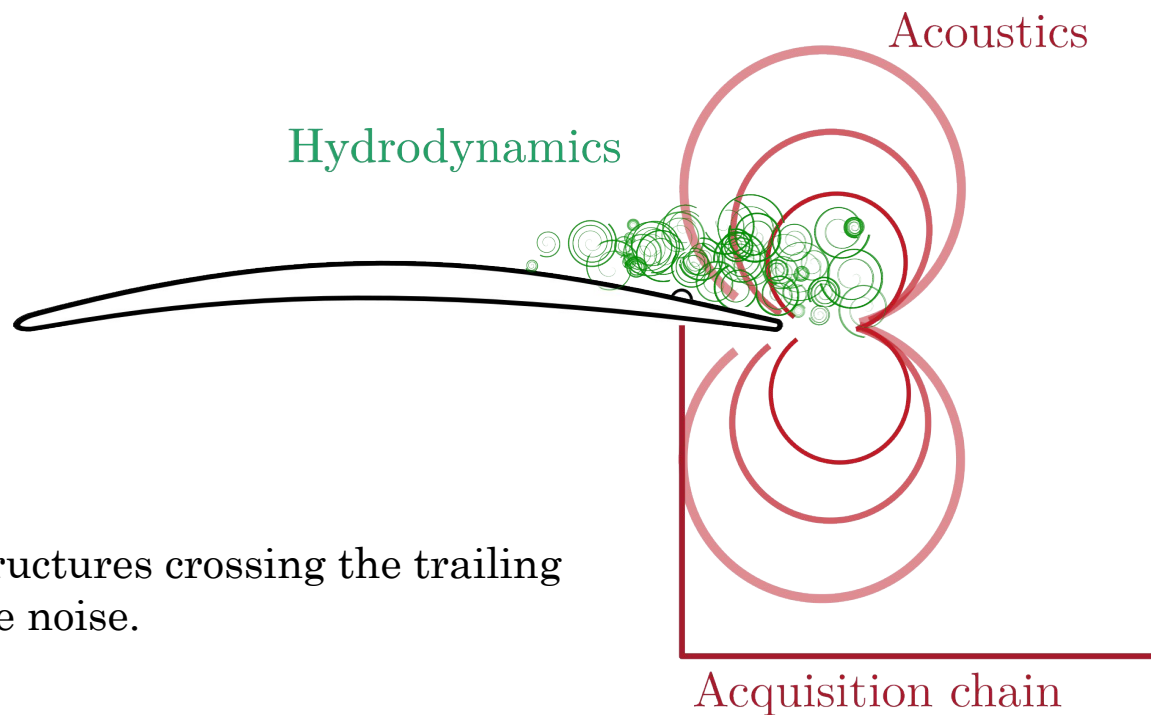


Analysis of Airfoil Self-Noise Sources Using Wavelet-Based Denoising

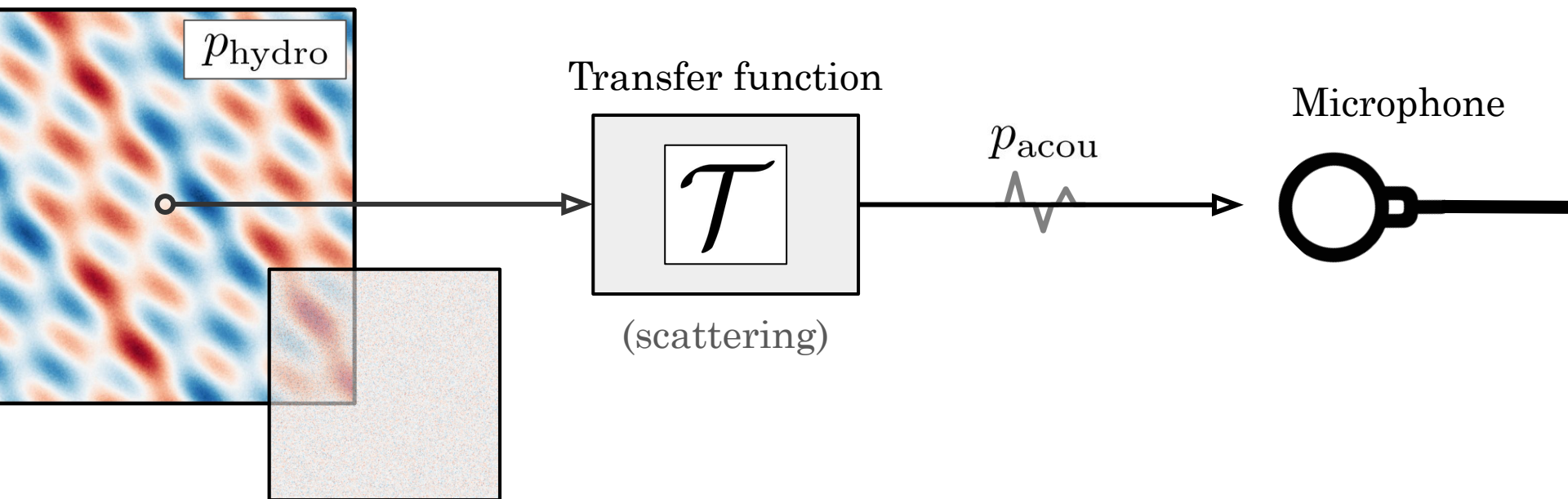
Enrico Foglia, Jose Manuel Rendon Arredondo, Stephane Moreau,
Michael Bauerheim, Thierry Jardin

Problem set up

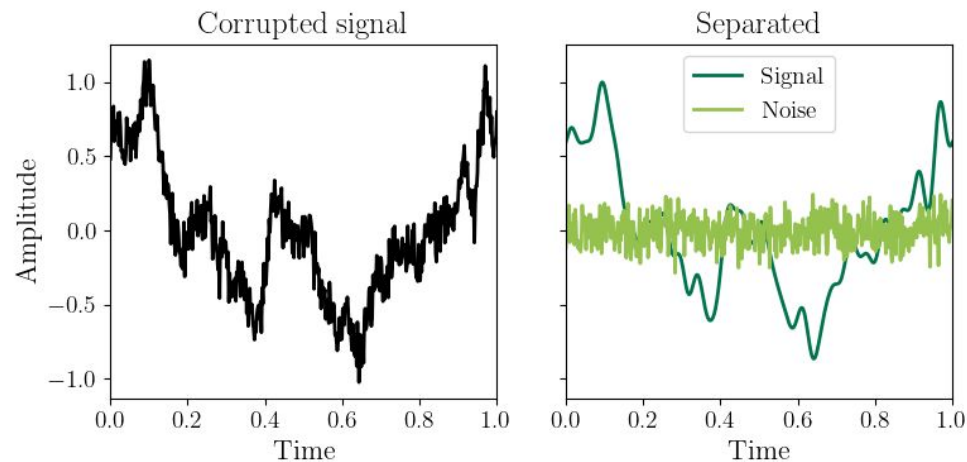


Turbulent structures crossing the trailing edge generate noise.

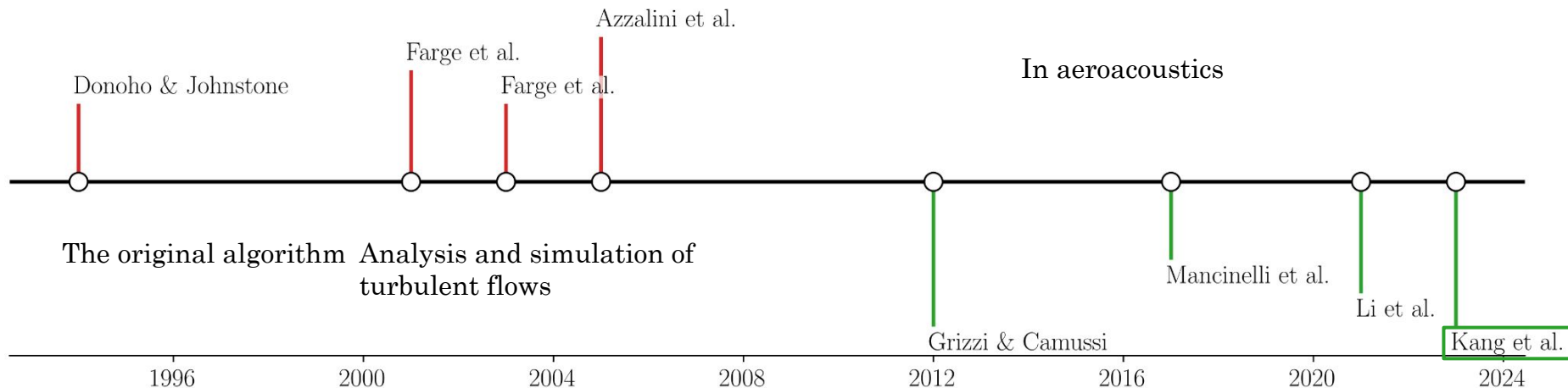
Problem set up



Why is it difficult?



Previous works



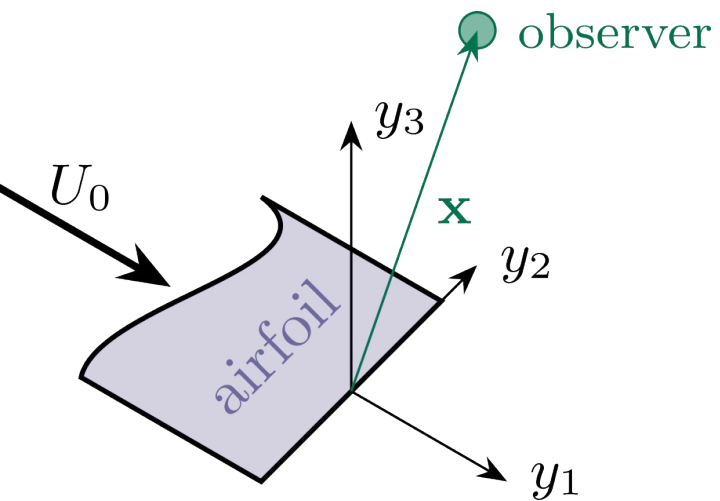
Do their results still hold on experimental data?

Can we say something more on the incoherent component?

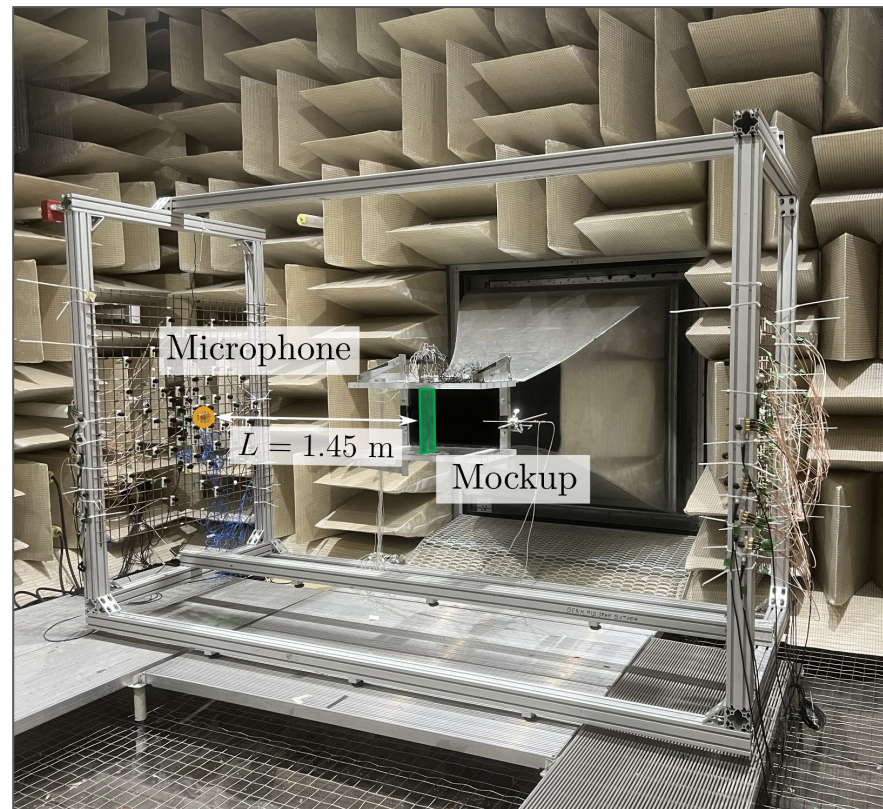
Content

- Trailing-edge noise data
 - Amiet's theory
 - Experimental set-up
- Wavelet-thresholding denoising
- Results
 - Correlation
 - Comparison with Wiener filter
 - Histogram

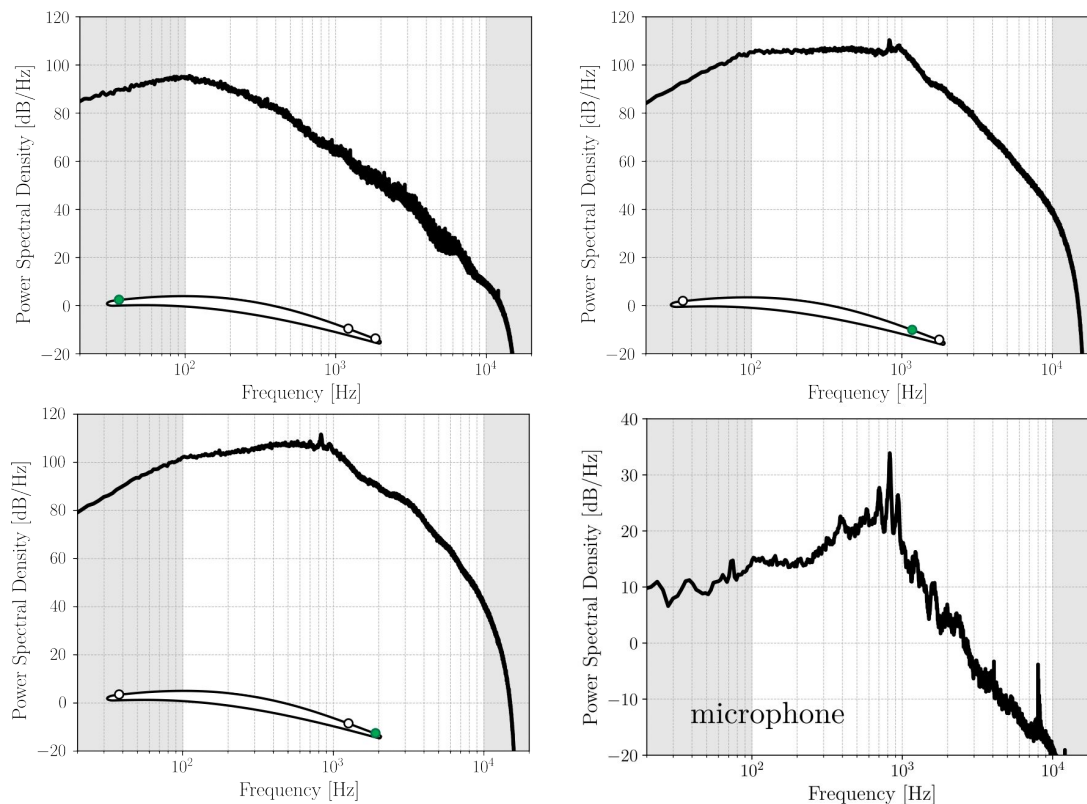
Amiet's theory



$$S(\omega, \mathbf{x}) = |\mathcal{T}(\omega, \mathbf{x})|^2 \underbrace{\Phi_{pp}(\omega) \ell_2(\omega)}_{\text{sources}}$$



Wall-pressure and far-field spectra

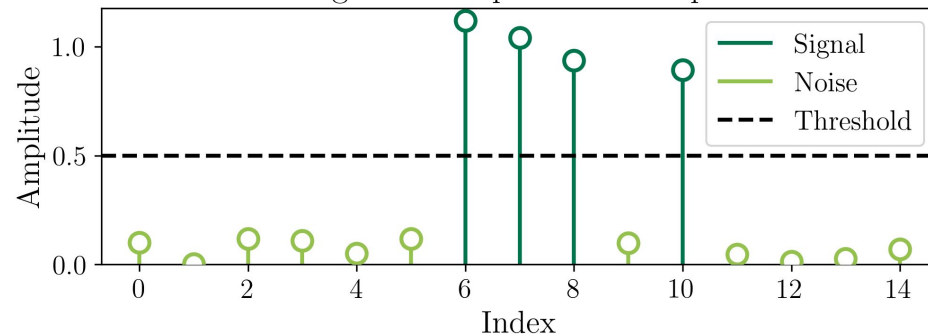


Wavelet-thresholding denoising

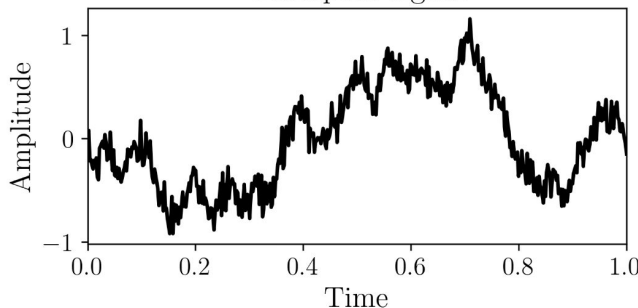
Decompose the signal into a basis:

$$f(t) = \sum_{j \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} \langle f, \psi_{j,n} \rangle \psi_{j,n}(t)$$

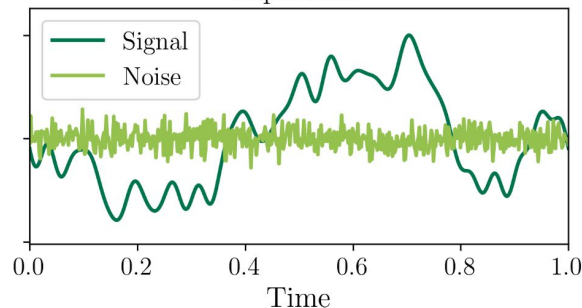
Signal decomposition example



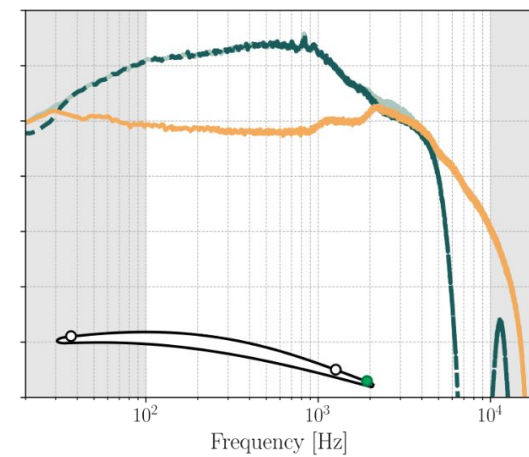
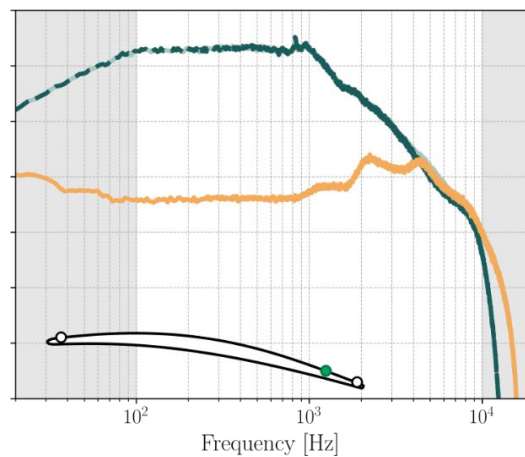
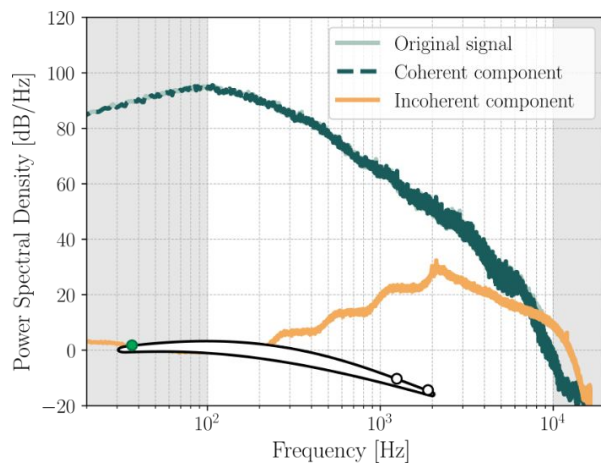
Corrupted signal



Separated

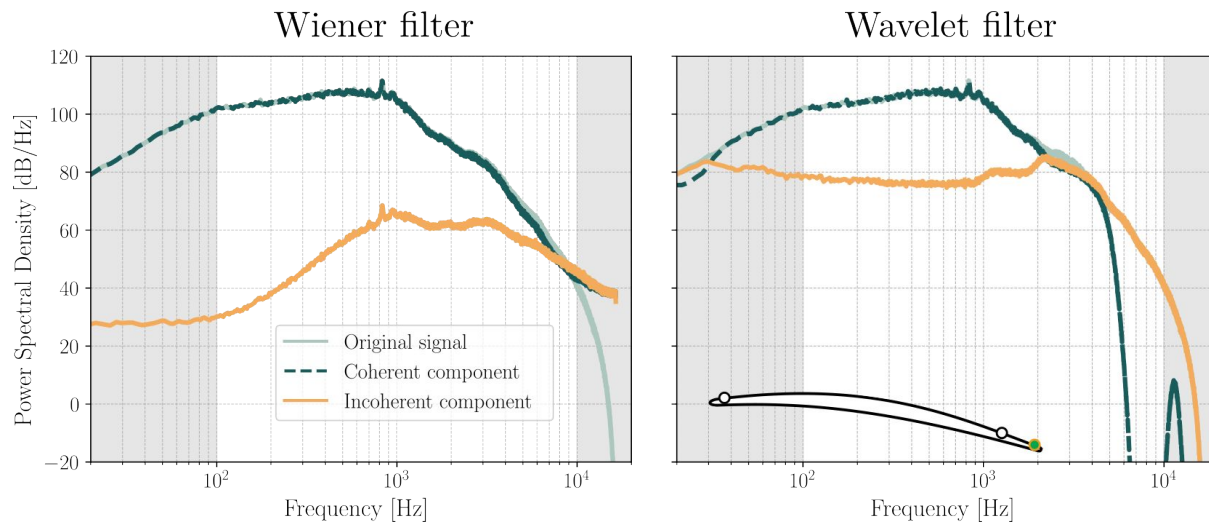


Filtered signal - PSD



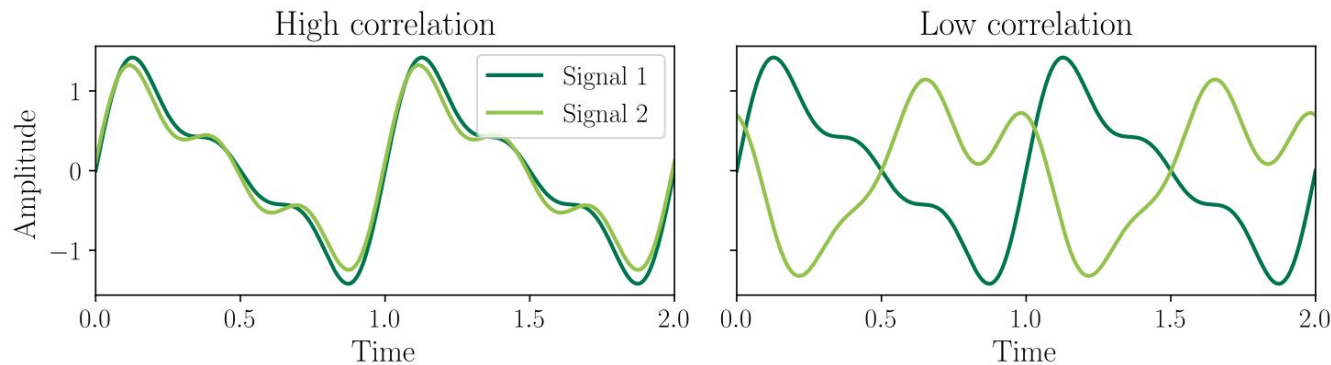
Comparison with Wiener filter (PSD)

Is there reason to prefer the wavelet filter?



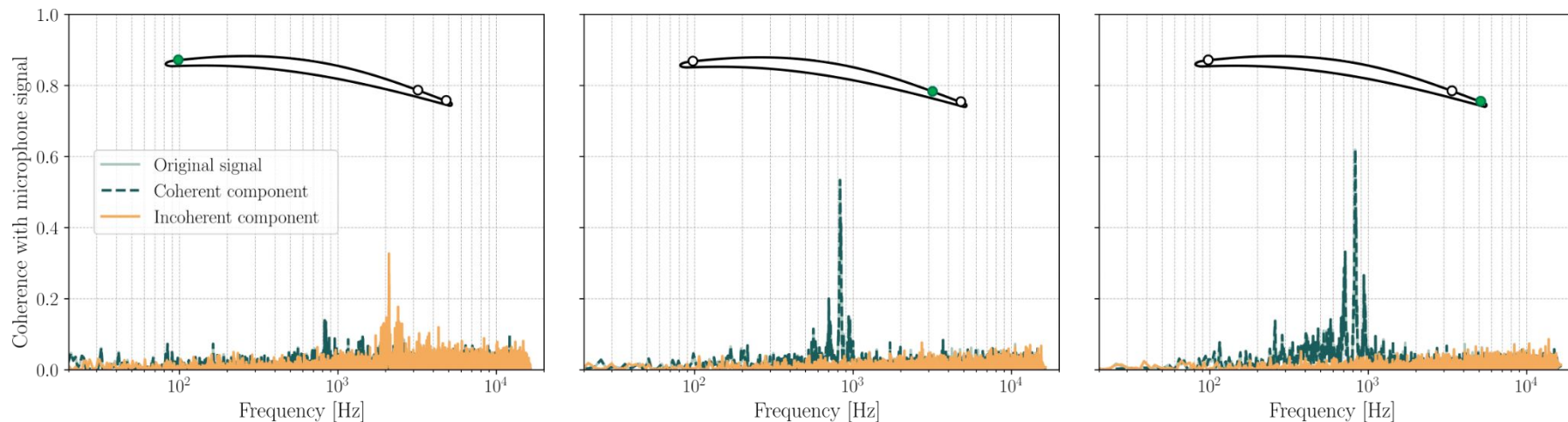
Coherence (definition)

$$\Gamma_{wm}(\omega) \triangleq \int_{\mathbb{R}} d\tau C_{wm}(\tau) e^{-i\tau\omega} = \int_{\mathbb{R}} d\tau \frac{\mathbb{E}\{p_{\text{wall}}(t)p_{\text{micro}}(t+\tau)\}}{\sqrt{\mathbb{V}\{p_{\text{wall}}\}}\sqrt{\mathbb{V}\{p_{\text{micro}}\}}} e^{-i\tau\omega}$$



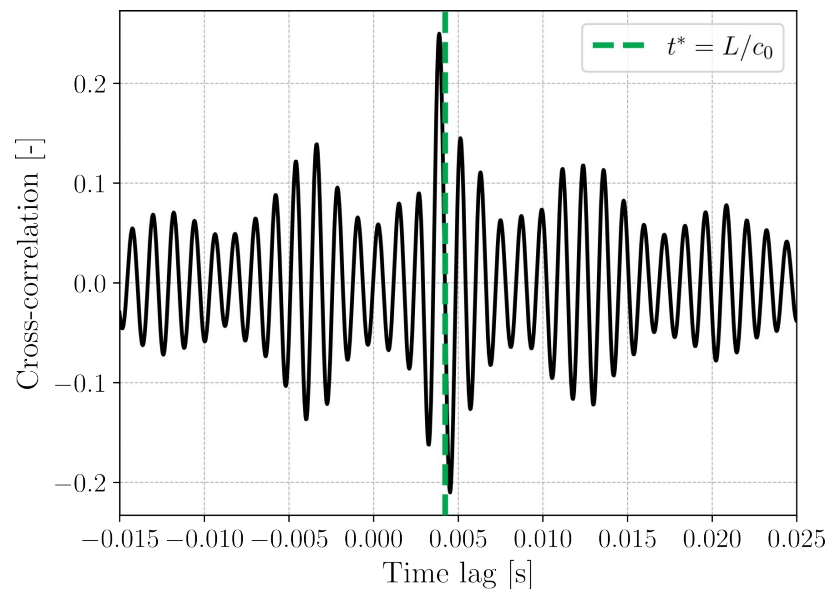
Coherence (results)

The only meaningful coherence is with the *coherent* component.



Correlation

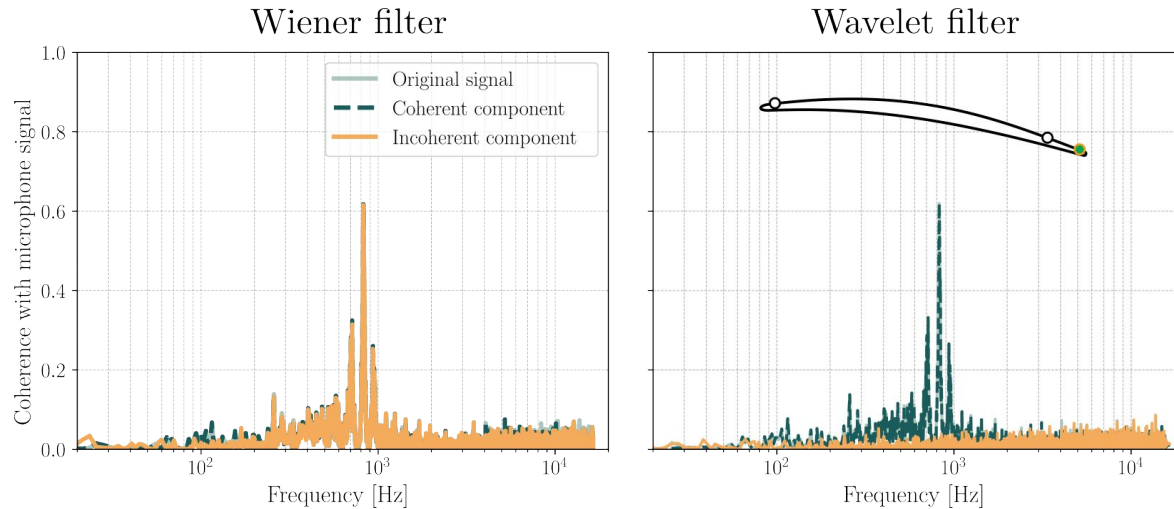
The maximum correlation is at the acoustic propagation time!



$$\frac{\mathbb{E}\{p_{\text{wall}}(t)p_{\text{micro}}(t + \tau)\}}{\sqrt{\mathbb{V}\{p_{\text{wall}}\}}\sqrt{\mathbb{V}\{p_{\text{micro}}\}}}$$

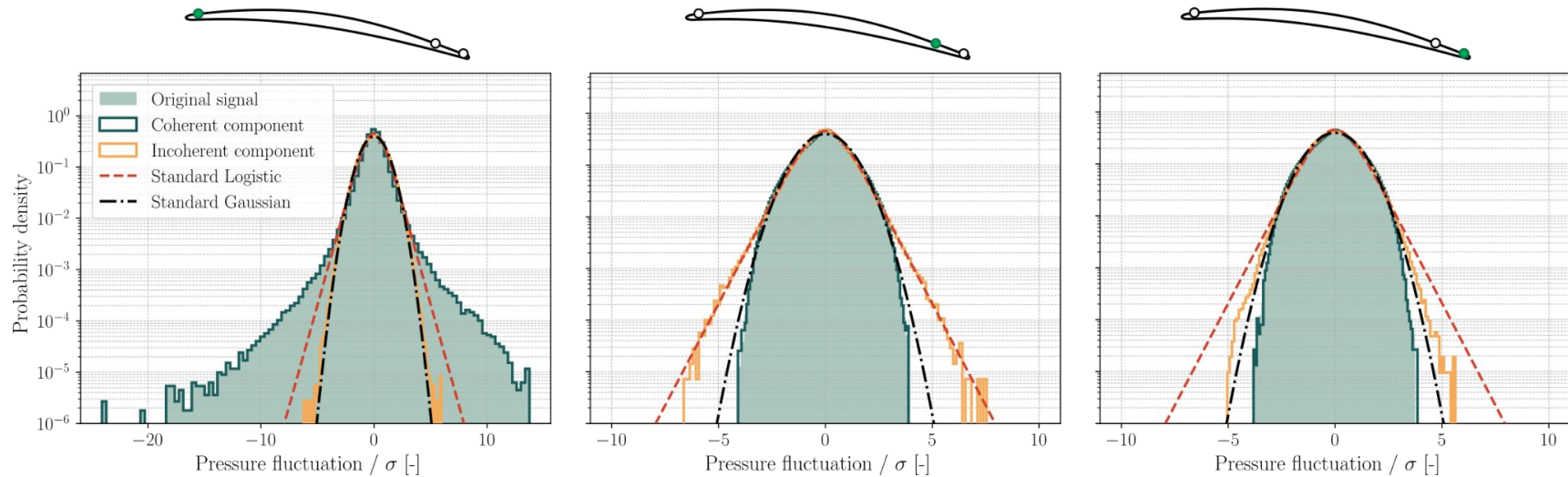
Comparison with Wiener filter (coherence)

The Wiener filter does not separate the signal into physically different components.

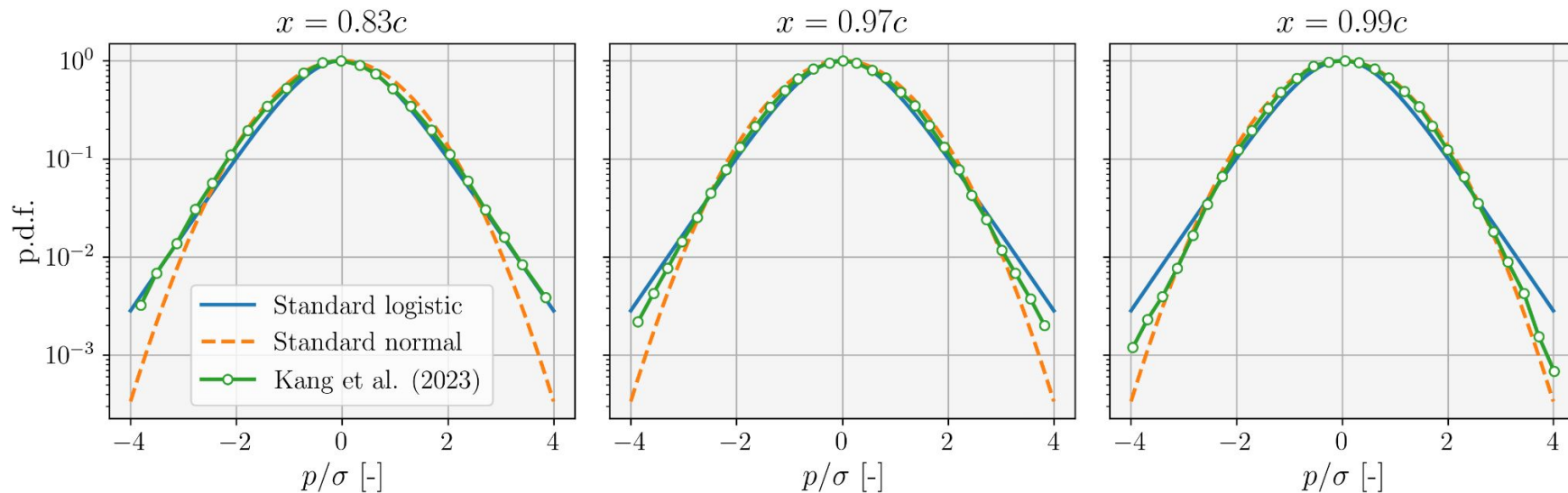


Histogram

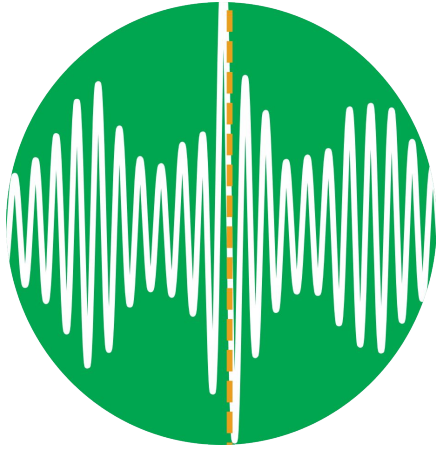
Logistic distribution in the fully turbulent zone. Back to Gaussian at the trailing edge.



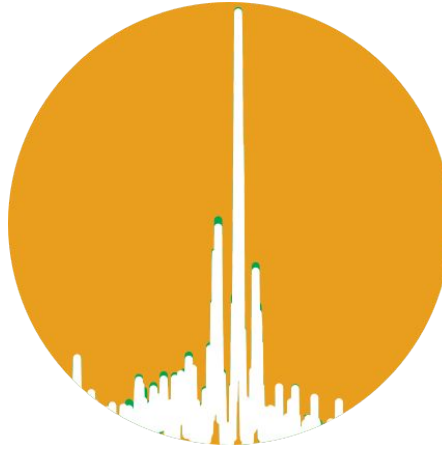
Kang et al. 2023: the logistic again



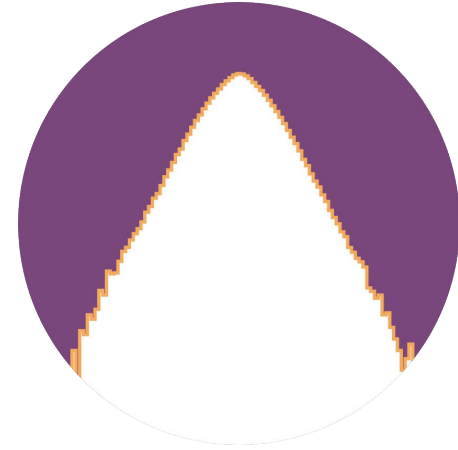
Conclusions



The **coherent component** is the only one correlated with the far-field sound. For this, we expect it to be mostly **hydrodynamics**



The Wiener filter is not able to perform a decomposition of the same kind as the wavelet one. **The filtering algorithm is important**



The histogram indicates that the **incoherent component comes from multiple sources** (acoustics, but also **turbulence!**)

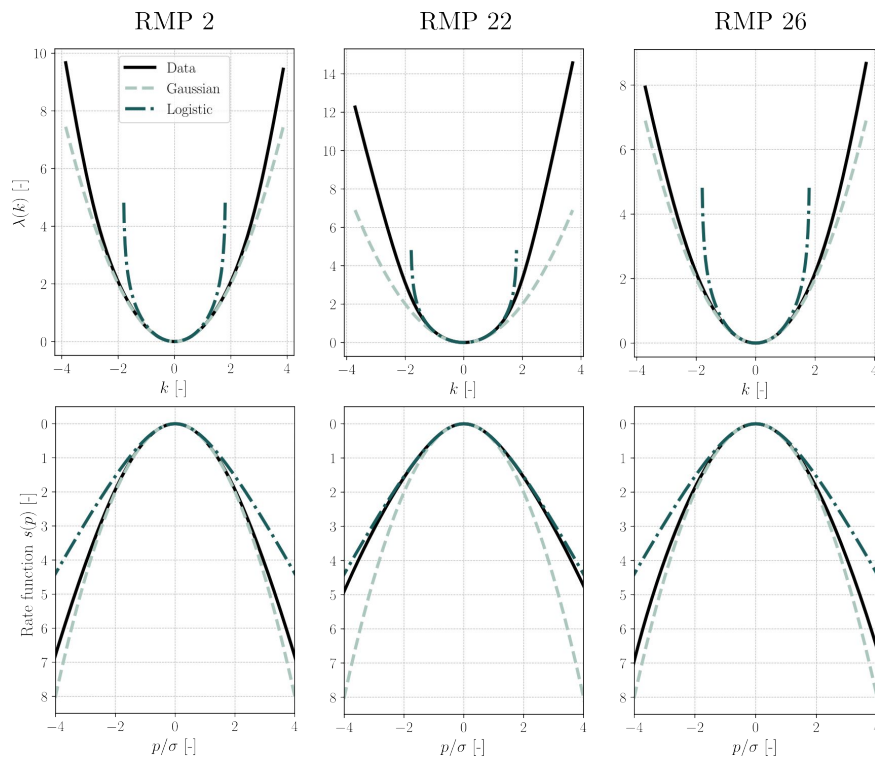
Perspectives

- Finalize analysis: compare Amiet complete signal vs denoised
- Repeat the analysis on compressible DNS
 - Get propagation speed
 - Problem: short time signals
- Why logistic?
 - Connect the linear tails of the histogram with theory of boundary layer turbulence (attached eddies? Extreme events?)

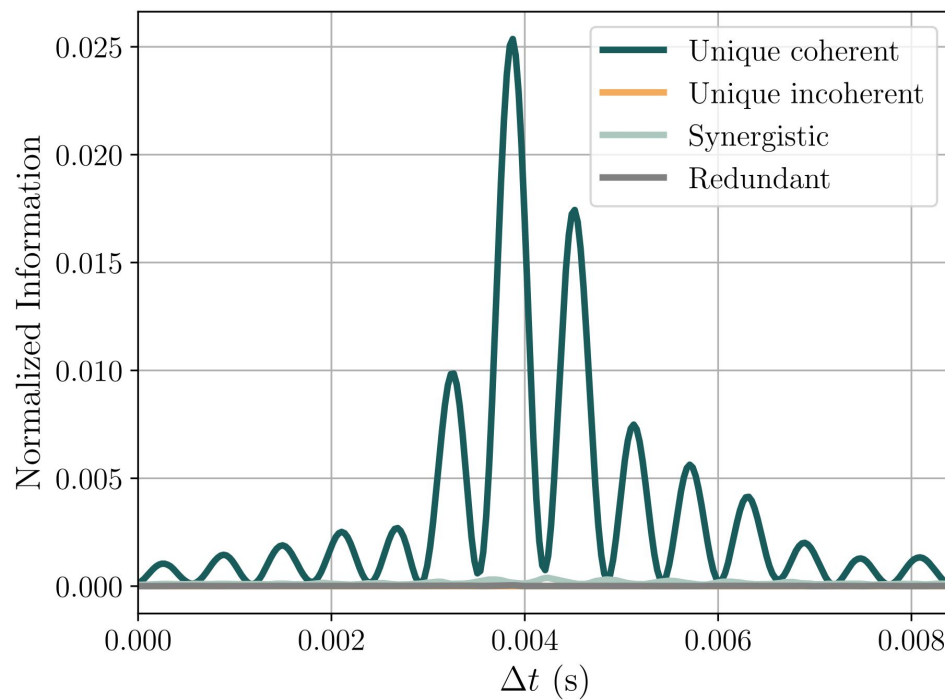


Thank you for your attention
Grazie per la vostra attenzione

LDT: results



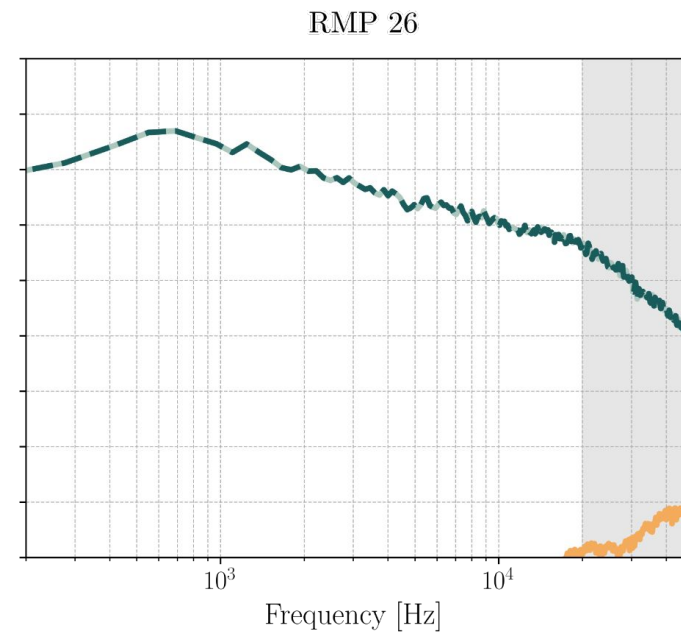
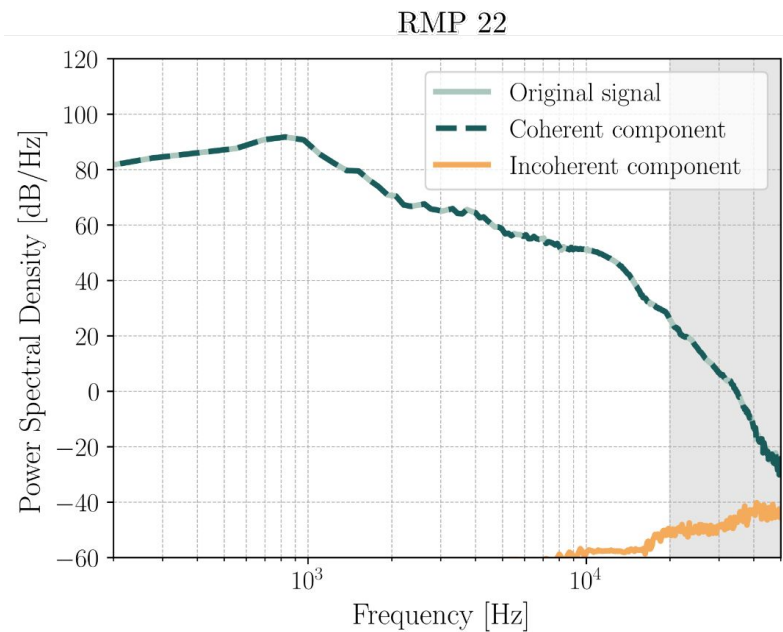
SURD: time lag



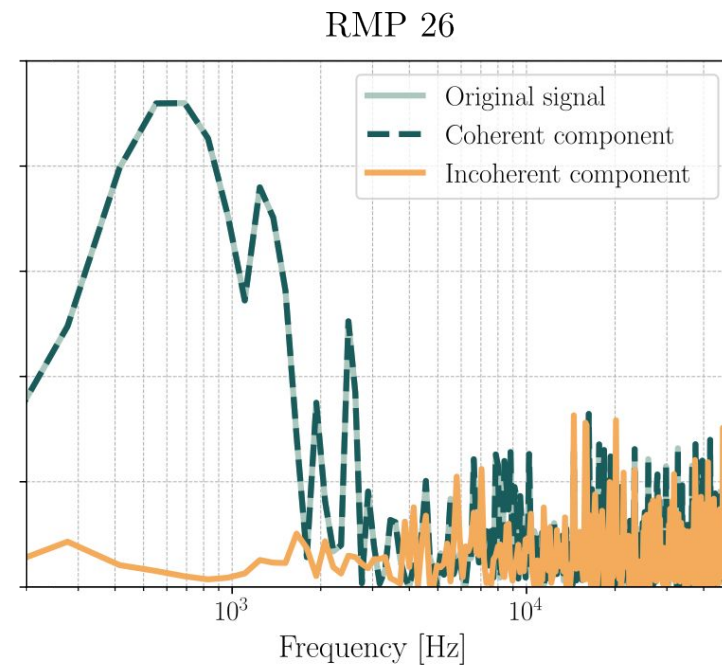
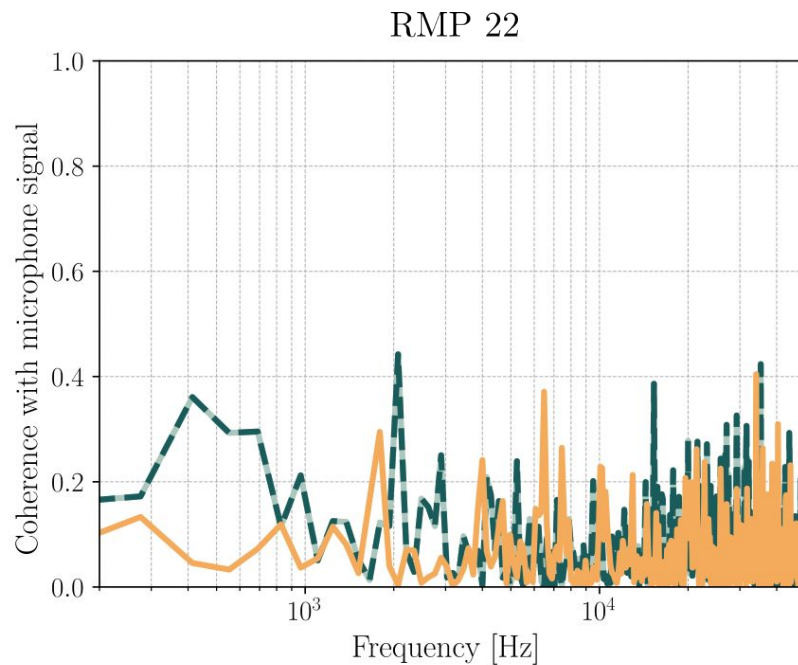
SURD: at optimal time lag

RMP	ΔI_{coh}^U	ΔI_{inc}^U	ΔI^R	ΔI^S	$H(Y \mathbf{X})$
2	0.9564	0.0000	0.0019	0.0417	99.98%
22	0.9984	0.0000	0.0002	0.0014	98.86%
26	0.9924	0.0000	0.0015	0.0061	97.44%

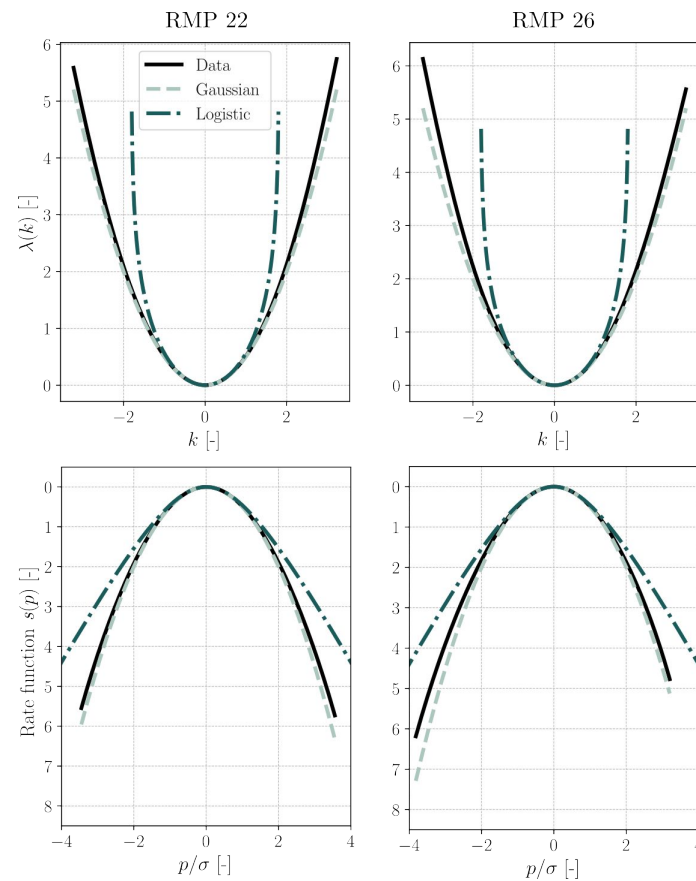
LBM: PSD



LBM: coherence



LBM: LDT



LBM: Wiener

